

Product Requirements Document (PRD)

Crime Analytics & Investigation Management Platform

1. Executive Summary

Product Vision

A secure, scalable, and analytics-driven law enforcement platform that centralizes criminal records, FIR management, investigations, evidence storage, officer activities, and crime intelligence reporting.

The platform aims to digitize investigation workflows while providing analytical insights that assist law enforcement agencies in decision-making and case management.

2. Business Objectives

Primary Goals

- Centralize crime-related data
- Improve investigation efficiency
- Reduce paperwork
- Provide crime intelligence dashboards
- Enable evidence management
- Improve officer accountability
- Support data-driven policing

Portfolio Goals

Demonstrate expertise in:

- Spring Boot Architecture
 - JWT Security
 - RBAC
 - REST API Design
 - Database Modeling
 - File Management
 - Analytics
 - Dashboard Development
 - Docker Deployment
-

3. Stakeholders

Role	Responsibility
Admin	System Management
Officer	FIR & Case Creation
Investigator	Investigation Activities
System Administrator	Deployment & Monitoring

4. User Roles & Permissions

Admin

Permissions:

- Create users
- Update users
- Delete users
- Assign roles
- View all records
- Manage officers
- View analytics
- Configure settings

Officer

Permissions:

- Create FIR
 - Create Criminal Profile
 - Open Case
 - Upload Evidence
 - Update Case Progress
-

Investigator

Permissions:

- View Assigned Cases
 - Add Notes
 - Update Investigation Status
 - Upload Evidence
 - Generate Reports
-

5. Functional Requirements

Module 1: Authentication & Authorization

Features

Registration

Fields:

- First Name
- Last Name
- Email
- Password

- Role

Validation:

- Email unique
 - Password minimum 8 characters
-

Login

Input:

```
{  
  "email": "john@example.com",  
  "password": "password123"  
}
```

Output:

```
{  
  "accessToken": "",  
  "refreshToken": "",  
  "role": "OFFICER"  
}
```

JWT Security

Flow:

Login

↓

Authenticate

↓

Generate Access Token

↓

Generate Refresh Token

↓

Return Tokens

↓

Access Protected APIs

Forgot Password

Workflow:

Email



Verification Link



Reset Password Page



Update Password

Module 2: Criminal Management

Criminal Profile

Fields

criminalId
firstName
lastName
gender
dob
nationality
address
photoUrl
status
riskLevel
createdAt

Status

- Wanted
 - Arrested
 - Released
 - Under Investigation
-

Features

- Create Criminal
- Update Criminal
- Delete Criminal

- View Criminal
 - Search Criminal
-

Module 3: FIR Management

FIR Fields

firId
firNumber
title
description
crimeType
incidentDate
location
status
createdBy
createdAt

Status

- Draft
 - Filed
 - Under Investigation
 - Closed
-

Features

- Create FIR
 - Update FIR
 - Search FIR
 - Change Status
-

Module 4: Case Management

Case Fields

caseId
caseNumber
title
description
priority
status
assignedOfficer
assignedInvestigator

Priority

- Low
- Medium
- High
- Critical

Status

- Open
- Assigned
- Investigating
- Suspended
- Closed

Features

- Create Case
 - Assign Case
 - Update Case
 - Close Case
-

Module 5: Evidence Management

Supported Files

- Images
- PDF
- Documents

Evidence Metadata

evidenceId
caseId
fileName
fileType
fileSize
uploadedBy
uploadedAt
description

Features

- Upload Evidence
 - Download Evidence
 - Search Evidence
 - Link Evidence To Case
-

Module 6: Investigation Management

Investigation Fields

investigationId
caseId
assignedInvestigator
status
startedAt
closedAt

Features

- Add Notes
 - Timeline Tracking
 - Update Status
 - Generate Closure Report
-

Module 7: Witness Management

Witness Fields

witnessId
firstName
lastName
phone
email
address
statement

Features

- Create Witness
 - Update Witness
 - Delete Witness
 - Search Witness
-

Module 8: Notification System

Email Notifications

Events:

- New Case Assignment
 - Status Changed
 - Evidence Uploaded
 - Investigation Reminder
-

Module 9: Crime Analytics

Dashboard Metrics

Crime Statistics

- Total Crimes
- Active Cases
- Closed Cases
- Solved Cases

Officer Metrics

- Cases Assigned
- Cases Closed
- Investigation Success Rate

Investigation Metrics

- Completion Rate
 - Average Resolution Time
-

Charts

Bar Chart

Crime Count By Type

Pie Chart

Solved vs Unsolved

Trend Graph

Monthly Crime Trends

Module 10: Search & Filtering

Global Search

Search Across:

- Criminals
- FIRs
- Cases

- Witnesses
 - Evidence
-

Filters

Crime Type
Date Range
Status
Location
Officer
Priority

Pagination

page
size
sortBy
direction

6. User Stories

Admin

US-001

As an Admin,
I want to create users,
So that officers can access the system.

Officer

US-002

As an Officer,
I want to create FIRs,
So that incidents are officially recorded.

Investigator

US-003

As an Investigator,
I want to add investigation notes,
So that case progress is documented.

Officer

US-004

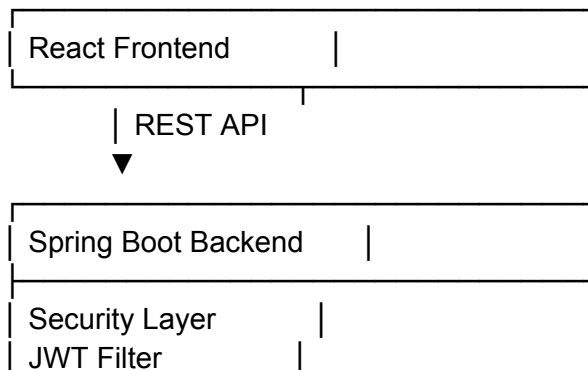
As an Officer,
I want to upload evidence,
So that proof can be attached to cases.

Admin

US-005

As an Admin,
I want analytics dashboards,
So that crime trends can be monitored.

7. System Architecture





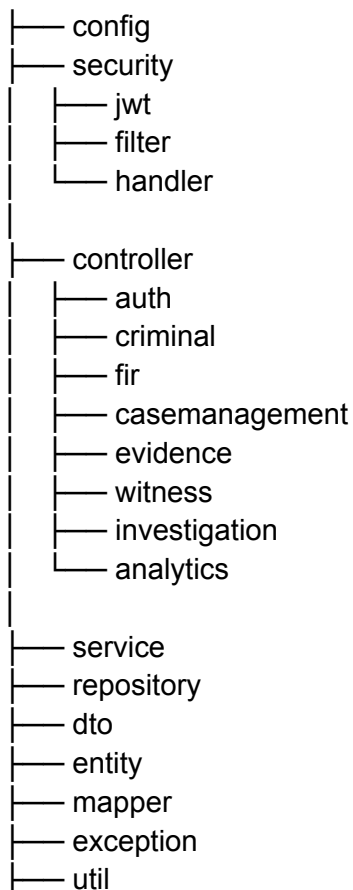
External Services

Google Maps API

Email Service

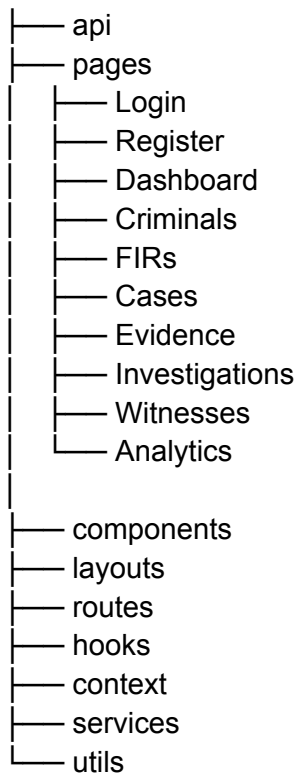
8. Backend Folder Structure

src/main/java/com/crimeanalytics



9. Frontend Folder Structure

src



10. Database Schema

USERS

users

id PK

first_name

last_name

email UNIQUE

password

role

enabled
created_at

OFFICERS

officers

id PK
badge_number UNIQUE
rank
department
user_id FK UNIQUE

CRIMINALS

criminals

criminal_id PK
first_name
last_name
gender
dob
address
status
risk_level
created_at

FIRS

firs

fir_id PK
fir_number UNIQUE
title
description
crime_type
location
status
officer_id FK
created_at

CASES

cases

case_id PK

case_number UNIQUE

title

description

priority

status

fir_id FK

assigned_officer_id FK

assigned_investigator_id FK

created_at

Remove

criminal_id FK

CASE_CRIMINAL

case_criminal

id PK

case_id FK

criminal_id FK

role_in_case

arrest_status

added_at

INVESTIGATIONS

investigations

investigation_id PK

case_id FK

status

started_at

closed_at

INVESTIGATION_NOTES

investigation_notes

note_id PK

investigation_id FK

author_id FK

note_text

created_at

updated_at

EVIDENCE

evidence

evidence_id PK

case_id FK

file_name

file_url

file_type

file_size

uploaded_by FK

uploaded_at

WITNESSES

witnesses

witness_id PK

first_name

last_name

phone

email

address

CASE_WITNESS

case_witness

id PK

case_id FK

witness_id FK

statement

NOTIFICATIONS

notifications

notification_id PK

user_id FK

title

message

status

created_at

Final ER Diagram

USER

|

| 1:N

|

OFFICER

USER

|

| 1:N

|

NOTIFICATION

OFFICER

|

| 1:N

|

FIR

FIR

|

| 1:N

|

CASE

OFFICER

|

| 1:N
|
CASE

CASE

|
| 1:N
|

INVESTIGATION

INVESTIGATION

|
| 1:N
|

INVESTIGATION_NOTE

USER

|
| 1:N
|

INVESTIGATION_NOTE

CASE

|
| 1:N
|

EVIDENCE

CASE

|
| N:M
|

WITNESS

CASE

|
| N:M
|

CRIMINAL

12. REST API Specification

Authentication

Method	Endpoint	Action
POST	/api/auth/register	Register
POST	/api/auth/login	Login
POST	/api/auth/refresh	Refresh Token
POST	/api/auth/forgot-password	Forgot Password

Criminal APIs

Create Criminal

POST /api/criminals

Get Criminal

GET /api/criminals/{id}

Update Criminal

PUT /api/criminals/{id}

Delete Criminal

DELETE /api/criminals/{id}

FIR APIs

POST /api/firs

GET /api/firs

GET /api/firs/{id}

PUT /api/firs/{id}

DELETE /api/firs/{id}

Case APIs

POST /api/cases
GET /api/cases
GET /api/cases/{id}
PUT /api/cases/{id}
DELETE /api/cases/{id}
POST /api/cases/{id}/assign

Evidence APIs

POST /api/evidence/upload
GET /api/evidence/{id}
GET /api/evidence/download/{id}

Analytics APIs

GET /api/analytics/dashboard
GET /api/analytics/crimes-by-city
GET /api/analytics/crimes-by-type
GET /api/analytics/monthly-trends
GET /api/analytics/officer-performance

13. Security Requirements

Authentication

- JWT Access Token
- Refresh Token
- Secure Logout

Password Security

- BCrypt Password Encoder

Authorization

ADMIN

Full Access

OFFICER

FIR + Criminal + Case

INVESTIGATOR

Assigned Cases Only

Additional Security

- Input Validation
 - SQL Injection Protection
 - XSS Protection
 - CSRF Disabled for JWT APIs
 - Rate Limiting
 - Audit Logging
-

14. Non-Functional Requirements

Performance

- API response < 500ms
- Dashboard load < 3 seconds

Scalability

- Stateless APIs
- Horizontal Scaling

Availability

- 99% uptime target

Logging

- Spring Boot Logging
- Audit Trail Logging

Monitoring

- Spring Boot Actuator
- Health Checks

Maintainability

- Layered Architecture
 - DTO Pattern
 - Global Exception Handling
-

15. Development Roadmap (60 Hours Total)

Week 1 (15 Hours)

Day 1

- Project setup
- GitHub repository

Day 2

- Database design

Day 3

- Entity creation

Day 4

- Repository layer

Day 5

- JWT Authentication

Day 6

- Role-based authorization

Day 7

- Swagger configuration
-

Week 2 (15 Hours)

Day 8

- Criminal module

Day 9

- FIR module

Day 10

- Case module

Day 11

- Witness module

Day 12

- Validation layer

Day 13

- Pagination & filtering

Day 14

- Testing
-

Week 3 (15 Hours)

Day 15

- Evidence upload

Day 16

- File storage

Day 17

- Investigation module

Day 18

- Notification service

Day 19

- Email integration

Day 20

- Maps integration

Day 21

- Testing
-

Week 4 (15 Hours)

Day 22

- Analytics queries

Day 23

- Dashboard APIs

Day 24

- React setup

Day 25

- Authentication UI

Day 26

- CRUD screens

Day 27

- Analytics dashboard

Day 28

- Integration testing

Day 29

- Dockerization

Day 30

- Documentation & deployment
-

16. Future Enhancements

Phase 2

- Facial Recognition Integration
- Criminal Network Analysis
- AI-Based Crime Prediction
- NLP FIR Summarization
- OCR for Evidence Documents
- Mobile Application
- Real-Time Notifications (WebSocket)

Phase 3

- GIS Crime Intelligence Dashboard

- Predictive Policing Models
 - Inter-Department Data Sharing
 - Blockchain Evidence Chain-of-Custody
 - ML-Based Officer Performance Analytics
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Final Scope Summary

This project includes:

- JWT Authentication & RBAC
- Criminal Management
- FIR Management
- Case Management
- Investigation Workflow
- Witness Management
- Evidence Storage
- Crime Analytics Dashboard
- Global Search & Filtering
- Email Notifications
- Maps Integration
- Swagger Documentation
- Docker Deployment
- React Frontend
- MySQL Database
- Enterprise-grade Spring Boot Architecture

This scope is large enough to be a strong portfolio project and demonstrate advanced Spring Boot backend engineering skills while remaining feasible within a ~60-hour implementation plan by focusing first on the MVP modules and expanding analytics and integrations incrementally.